

Farzaneh Fayazbakhsh

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Personal Statement

Interdisciplinary researcher combining bioinformatics, genomics and molecular biology, with hands-on experience in sequencing workflows, cell culture and translational research. I use R, Python, Linux and reproducible pipelines to analyze high-dimensional biological data, including single-cell RNA-seq and spatial transcriptomics, and I am motivated to investigate how gene-regulatory programs and tissue context shape developmental disease mechanisms.

Education

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| M.Sc. Swedish University of Agricultural Sciences , Bioinformatics | 2025–present |
| <ul style="list-style-type: none"> Highest grade (5/5) in Bioinformatics course Thesis: Exploring the Potential of Fragile Sites as Markers for Lineage Tracing | |
| KTH Royal Institute of Technology , Single-Cell & Spatial Transcriptomics Data Analysis Course | 2024 |
| M.Sc. Tehran University of Medical Sciences , Medical Nanotechnology | Iran |
| <ul style="list-style-type: none"> Awarded as the Top Master's Graduate (GPA: 4/4) Thesis: Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip | 2017–2022 |
| B.Sc. Alzahra University , Cellular and Molecular Biology (Biotechnology Specialization) | Iran |
| | 2012–2016 |

Bioinformatics & Computational Biology Projects

Automated Raw Sequencing Data Quality Control Pipeline

- Built an automated, parallelized Bash orchestration workflow designed for execution on High-Performance Computing (HPC) clusters via SLURM scheduling.
- Deployed FastQC and fastp dynamically inside isolated, version-controlled Singularity and Apptainer containers to automate raw sequence pairing, quality checking, and adapter trimming.
- Implemented multi-core processor allocation inside container runtimes to optimize throughput when executing string substitutions and batch file calculations.
- Orchestrated MultiQC container tasks to aggregate and output comprehensive data quality matrices, tracking processing adjustments across all sample cohorts simultaneously.

Tumor Microenvironment Profiling: scRNA-Seq & Spatial Transcriptomics

- Engineered end-to-end transcriptomic pipelines in R using Seurat and SCTransform to map high-dimensional gene expression across breast cancer tissue cohorts.
- Executed data pre-processing, Quality Control (QC), linear (PCA), and non-linear (t-SNE / UMAP) dimensionality reduction alongside graph-based clustering to resolve distinct cell populations and structural tissue features.
- Containerized both analysis environments using Docker and Docker Compose, ensuring full cross-platform compatibility and seamless runtime execution across varied infrastructure platforms.
- Managed pipeline documentation, software repositories, and strict workflow reproducibility utilizing Git and GitHub version control standards.

Research Experience

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| Karolinska Institute & SciLifeLab , Master's Thesis Research | 2026–present |
| <ul style="list-style-type: none"> Investigating replication-stress-induced genomic instability at common fragile sites using targeted long-read sequencing. Performing neuroblastoma cell culture, drug treatments, DNA extraction, and long-range PCR. Developed bioinformatics workflows using Python, Linux and HPC environments for long-read sequencing analysis. Performed structural variant discovery, breakpoint characterization and endogenous barcode identification. | |
| KTH Royal Institute of Technology & SciLifeLab , Research Engineer | Sweden |
| <ul style="list-style-type: none"> Independently developed Tumor-on-a-Chip models for in vitro anti-cancer drug screening. | 2023–2024 |

- Optimized layer-by-layer cellulose nanofibril coatings to enhance cancer spheroid formation.
- Contributed to projects on Brain-on-a-Chip models for metabolic studies.
- Functionalized electrochemical sensors for ketone detection.

Tehran University of Medical Sciences, Master's Thesis Research

Iran
2018–2022

- Synthesized gold nanoparticles conjugated with polyphenols.
- Developed microfluidic devices for endothelial cell culture.
- Modeled early-stage endothelial dysfunction and conducted oxidative stress assays.
- Assisted in preparing a PET/PDMS/Collagen nanofibrous membrane for microfluidic cell culture.

Gen-Iran Lab, Pathology and Cytology Intern

Tehran, Iran
2023

- Hands-on experience in tissue preparation, fixation, embedding, and sectioning.
- Performed histological and immunohistochemical staining.

Computational Biology & Bioinformatics Skills

Programming & Pipeline Automation: Python, R, UNIX/Linux Shell, Nextflow (pipeline development), Git, and GitHub for version control

Genomics, High-Throughput Sequencing & Genetics: Practical experience in NGS data analysis, Whole Genome Sequencing (WGS), RNA-Seq processing, GWAS, QTL analysis, gene mapping, structural variation, metagenomics, genome annotation, and phylogenomics (MEGA, Clustal Omega)

Structural Biology, Modeling & Machine Learning: Sequence analysis, primer design, AlphaFold (protein structure prediction), PyMOL (molecular visualization), and foundational familiarity with deep learning frameworks

Infrastructure, HPC & DevOps: Hands-on experience with High-Performance Computing (HPC) clusters, containerization (Docker, Docker Compose, Apptainer/Singularity), and implementation of FAIR Research Data Principles

Wet Lab & Experimental Expertise

Cell Culture & Cancer Biology: 2D & 3D cell culture, tumor spheroid formation and organoid culture, in vitro cancer biology drug screening

Molecular Biology & Genomics Workflows: DNA extraction, downstream cellular and molecular assessments, PCR / Real-Time PCR, agarose gel electrophoresis, Agilent TapeStation analysis, flow cytometry, fluorescent microscopy

Microfluidics & Biosensors: Organ-on-a-Chip modeling, PDMS-based microfluidic systems fabrication, surface and electrochemical sensor functionalization

Histology & Pathology: Histological sample preparation, H&E staining, immunohistochemistry (IHC)

Nanotechnology & Nanomaterial Synthesis: Green synthesis of gold nanoparticles, nanofiber electrospinning, cellulose nanofibrils (CNF) coating optimization

Peer-Reviewed Publications

Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip 2023

Farzaneh Fayazbakhsh, Fatemeh Hataminia, Houra Mobaleghol Eslam, Mohammad Ajoudanian, Sharmin Kharrazi, Kazem Sharifi, Hossein Ghanbari
Scientific Reports

Metabolic Assessment of iPSC-Derived Neurons Under Ketogenic and Normal Diets: Ketone Sensor Development and Comparative Analysis of Neuronal Metabolism 2026

Rohollah Nasiri*, **Farzaneh Fayazbakhsh***, Reyhaneh Sanei, Tingting Wu, Nayere Taebnia, Rouhollah Habibbey, Omid Fotouhi, John Cognetti, Volker M. Lauschke, Aman Russom, Anna Herland
CellPress

Assessing Layer-by-Layer Assembly of Cellulose Nanofibrils in Pancreatic Tumor Spheroid Formation 2023

Negar Abbasi Aval, Ekeram Lahchaichi, Oana Tudoran, **Farzaneh Fayazbakhsh**, Rainer Heuchel, Matthias Löhr, Torbjörn Pettersson, Aman Russom
Biomedicines

Characterization and Numerical Simulation of a New Microfluidic Device for Cell Investigations	2024
H Meslam, F Hataminia, H Asadi-Saghandi, F Fayazbakhsh , N Tabatabaei, Hossein Ghanbari	
Frontiers in Lab on a Chip Technologies	
Nanofibers in Respiratory Masks: An Alternative to Prevent Pathogen Transmission	2022
Tabatabaei SN, Faridi-Majidi R, Boroumand S, Norouz F, Rahmani M, Rezaie F, Fayazbakhsh F , Faridi-Majidi R	
IEEE Transactions on NanoBioscience	

Conference Presentations

Spheroid Formation in Cellulose Nanofibrils-Coated Microfluidic Chips	μTAS 2024, Mon-
Poster presentation (first author)	treal, Canada
	2024
Design of a Microfluidic Electrochemical Ketone Sensor	Swedish Mi-
Oral presentation (co-author)	crofluidics in
	Life Science,
	KTH, Sweden
	2023
Investigation of Effect of Different Diets on Brain Energy Metabolism Using Organ-on-a-Chip	MPS World Sum-
Poster presentation (co-author)	mit, Berlin, Ger-
	many
	2023
Preparation of PET/PDMS/Collagen Nanofibrous Membrane Embedded in a Microfluidic Device for Organ-on-Chip Applications	The 3rd Con-
Poster presentation (co-author)	ference on
	Nanofibers,
	Tehran Univer-
	sity of Medical
	Sciences
	2022

Professional & Personal Skills

Bridging Wet & Dry Lab: Confident working in both experimental labs and computational environments, making it easy to translate biological questions into data-driven answers

Independent Problem Solver: Highly autonomous in research environments, with a proven ability to troubleshoot bottlenecks and quickly teach myself new tools to keep projects moving forward

Collaborative Team Player: Experienced in international, interdisciplinary teams such as KTH, Karolinska Institute and SciLifeLab, with a strong focus on knowledge-sharing and helping colleagues

Clear Communicator & Fast Learner: Skilled at explaining complex data clearly through peer-reviewed publications and conference presentations; highly adaptable and quick to pick up new scientific topics and skills

Teaching & Mentorship

KTH Royal Institute of Technology , Intern Supervisor	Sweden
Guided an intern in laboratory techniques.	
Volunteer Service , Science Teacher	
Taught science to underprivileged children.	

Awards & Honors

Top Master's Graduate	Tehran Univer-
	sity of Medical
	Sciences
	2023
LUSH Prize Candidate	2022
Shortlisted for the Young Researcher Award.	

Darolfonoon Student Award

2020

For voluntary assistance in nanofiber mask production during the COVID-19 pandemic.

Top Article Award

Prize winner.

First National
and Student
Conference on
Applied Biology,
Alzahra Univer-
sity, Iran
2014

Top 5% in Iran's Konkour for Master's Degree University Entrance

Iran

Ranked 37th in Iran's national Konkour for the Master's Degree University Entrance Exam.

2016

References

Dr. Kasper Karlsson #linebreak() Department of Oncology-Pathology, Karolinska Institute #linebreak() kasper.karlsson@ki.se



Dr. Erik Bongcam Rudloff #linebreak() Department of Animal Biosciences, Swedish University of Agricultural Sciences #linebreak() erik.bongcam@slu.se ▼

Dr. Aman Russom #linebreak() Nanobiotechnology Division, Protein Science Department, KTH Royal Institute of Technology #linebreak() aman.russom@scilifelab.se ▼

Cheng-de (Ernest) Liu #linebreak() Department of Oncology-Pathology, Karolinska Institute #linebreak() chengde.liu@ki.se ▼

Dr. Anna Herland #linebreak() Nanobiotechnology Division, Protein Science Department, KTH Royal Institute of Technology #linebreak() aherland@kth.se ▼

Dr. Rohollah Nasiri #linebreak() Radiation Oncology Department, Stanford University #linebreak() rhnasiri@stanford.edu ▼